CHAPTER 1



THE EARLY YEARS

Tracking, in the context of spaceflight, refers to the collection of spacecraft position and velocity measurements so that its motion on orbit may be determined. The roots of modern day tracking and data acquisition can be found in the immediate years following the Second World War when the United States entered into a period of intense research to develop a viable ballistic missile technology. Science, and in particular, the development of sounding rockets (small rockets launched into the upper part of the atmosphere for research and experiment), played a part in this advancement. However, its pace was clearly driven by national security.

World War II had just forced the United States and the Soviet Union to become reluctant allies. With the war now over, the potential for an armed conflict between the two superpowers was something that both governments took careful steps to avoid while at the same time, and in no uncertain terms, prepared for. Even in the last days of the war, both countries postured to shape the political landscape that would soon emerge after Nazi Germany and the Imperial Japanese governments had been defeated. It was in this Cold War atmosphere that a major concern arose in the American military leadership, one that fueled the perception that the United States not only trailed, but trailed badly, the Soviet Union in the area of ballistic mis-

sile technology.[1] Much of this concern could be traced to that fact that much larger missiles were then needed by the Soviets to carry their nuclear warheads which were, at the time, bigger and much heavier than their American counterparts.[2] Also capitalizing on this state of apprehension was the very effective use of propaganda by the Soviets, as the usually secretive society openly and routinely paraded these large rockets in Red Square during times of state celebrations.

As the pace of missile testing accelerated, the Department of Defense (DOD) converted several former War Department weapon proving grounds in the continental United States into missile ranges. Key among these was the White Sands Missile Range (WSMR) in southern New Mexico. Here, captured German V-2 rockets were brought and flown after the war. In a move called Operation Paperclip, German rocket scientists, under the direction of Wernher von Braun, brought over 300 boxcar loads of missile materials to White Sands where they were tested as sounding rockets equipped with atmospheric sampling devices and telemetry transmitters.[3] It was in New Mexico where America's first suborbital rockets were launched above Earth's atmosphere in 1946 when the WAC Corporal ventured some 80 kilometers (50 miles) up, almost into the ionosphere.[4]

Other pioneering work at the Army range included the "Albert monkey flights" and the Navy's Aerobee sounding rockets.[5] In 1950, the Air Force converted the Long Range Proving Ground at Cape Canaveral, Florida into the Atlantic Missile Range, which it later renamed the Eastern Test Range (ETR). With a corridor that extended all the way from the Florida coast, over the Caribbean to Ascension Island in the South Atlantic, ETR was perfect for testing the early long-range intercontinental ballistic missiles (ICBM) in the Air Force arsenal such as the venerable Atlas. Located on the coast of Florida, easterly launches out of "The Cape" also enjoyed the enormous benefit of having the velocity from Earth's rotation "kick start" a rocket's journey into space. On the other coast, the Navy established the Pacific Missile Range (PMR), tracking missiles launched out of Point Mugu, California and later, Air Force launches from Vandenberg Air Force Base, on intercontinental trajectories to Kwajalein in the Marshall Islands some 10,000 kilometers (6,200 miles) away in the South Pacific.

In those days, there were essentially two proven methods for tracking vehicles into the upper atmosphere: optical and radar. *Optical tracking* has been around since humans first studied the stars and sailed the seas. As the name implies, visual sightings from ground-based, high-power telescopes were used to provide measurements of a satellite's position against the background stars. In April 1956, the Working Group on Tracking and Computation (WGTC), as part of the National Academy of Sciences' Space Science Board, approved a plan by the Smithsonian Astrophysical Observatory (SAO) in

Cambridge, Massachusetts, to establish an optical tracking network to photograph very small objects in anticipation of tracking the International Geophysical Year (IGY) satellites. The IGY was an 18-month (July 1957 through December 1958) scientific undertaking sponsored by the International Council of Scientific Unions designed to promote and stimulate a broad, worldwide investigation of Earth and the near-Earth cosmic environment. At the time, it was quite the watershed event in terms of fostering scientific interest on an international basis when East-West tensions were dominating the news.

To prepare for U.S. IGY activities, the SAO was given $3,380,000 by the National Academy of Sciences (NAS) to implement an optical network. This network, proposed by Harvard astronomer Fred L. Whipple and Ohio State University professor J. Allen Hynek, featured sophisticated Baker-Nunn high precision telescopic cameras that had unusually large apertures. These high-resolution cameras were named after its principal creators, James G. Baker, consultant to the Perkin-Elmer Corporation of Norwalk, Connecticut, and Joseph Nunn of South Pasadena, California. Built by the Boller and Chivens Company of South Pasadena and instrumented with optics fabricated by Perkin-Elmer, the Baker-Nunn was basically an f/1 Schmidt camera with a 20-inch (50.8-centimeter) focal length and a wide 5° by 30° field-of-view so as to accommodate star-field photography for the purposes of optical tracking. (A Schmidt camera is a high-powered telescope with a film plate holder instead of eyepiece at its focus. It is typically used as a survey instrument in which a large amount of sky is covered.) Another key component of the camera was its film. Manufactured by the Kodak Corporation, film for the Baker-Nunn used high speed Royal X-Pan emulsion, standard for high grain black and white resolution at the time. The 55.625-mm film size was, however, unique to the needs of the SAO. To achieve the required resolution, a single frame was rather large, measuring 2.2 inches by 18 inches (5.6 centimeters by 45.7 centimeters). It required close to an hour to process and dry the film in preparation for making position measurements.[6]

The camera operated by alternately tracking a satellite and the star background. Superimposed on the same strip of film was the image of a crystal-controlled clock from a separate optical system which was periodically illuminated by strobe lights. This was done to establish a precise time reference. From the photographs, the position of a satellite could be accurately determined by comparison with the position of the background star field.[7] The instrument was sensitive enough; under favorable lighting conditions, it could photograph a 16th magnitude object, which corresponds to something that is 25 times dimmer than the faintest star visible to the naked eye.[8]

By the late 1950s, the Smithsonian's optical network was global in scope and was concentrated in a geographical band of about 40° north and

south of the Equator. There were 12 stations located at sites selected, among other things, for their year-round mild weather:

North America
  Haleakala, Maui
  Jupiter, Florida
  Organ, New Mexico
South America
  Arequipa, Peru
  Curacao, Dutch Antilles
  Villa Dolores, Argentina
Australia
  Woomera
Africa
  Olifantsfontein, South Africa
Europe
  San Fernando, Spain
Asia
  Mitaka, Japan
  Naini Tal, India
  Shiraz, Iran[9]

Although optical tracking could be effective, it had its drawbacks. In particular, its usefulness was limited by something known as the acquisition problem. Simply put, in optical tracking, there was no way to initially find what it was supposed to track. Before the optical network could be put to use, the SAO had to come up with a plan to address the acquisition problem. Here, Whipple had an idea. To encourage participation and popular interest in this new field, Whipple's plan was to use a network of amateur volunteers worldwide to literally serve as eyes to visually find—or acquire—a satellite. The strategy was rather simple. After surveying the night sky and finding the satellite, these participants would pass the "acquisition data" on to one of the 12 Baker-Nunn stations that was closest to them.

Not surprising, the intention to use amateurs in this fashion was initially ill-received by the NAS Space Science Board's technical working group who thought the work "too technical" for just anyone to perform. But just as Whipple had anticipated, the announcement of the formation of Operation Moonwatch in early 1956 brought an enormous response. Visual observation teams sprang up in North and South America, Africa, Europe, Asia, in the Middle East and even at such remote places as Station C and Fletcher's Ice Island T-3 in the Arctic Basin. In the Washington, DC area, the first Moonwatch station equipped with 12 telescopes was set up in an



An optical satellite tracking network using Baker-Nunn cameras established by the Smithsonian Astrophysical Observatory was in place by the late 1950s. The camera would alternately track a satellite and the star background to obtain a fix on its location. Baker-Nunn cameras were also used extensively by the Air Force to track military satellites. (*http://history.nasa.gov/SP-4202/p9-151.jpg*)

apple orchard in Colesville, Maryland. The orchard was owned by a Mr. G. R. Wright, a Weather Bureau employee who ended up chairing the national advisory committee in charge of planning the unusual network.

Dry runs to test the ability of spotters to actually sight a tiny sphere hurling through space began in June of 1957. An imitation of the proposed IGY satellite flew over Fort Belvoir, Virginia. This test satellite was definitively "lo-tech," a makeshift device made up of a tiny flashlight bulb fastened to a plumber's suction plunger that was towed at the end of a clothesline behind an airplane flying at 1800 meters (6,000 feet). Light output from this object was calculated by Whipple to be about the same as that reflected off a metal sphere on orbit from a rising or setting Sun. The contraption worked though, and the experiment went off without a hitch. Soon, variations of the technique were used to train Moonwatchers from around the world. Two years later, some 250 teams with approximately 8,000 members were functioning under Moonwatch. Teams came from all walks of life, organized by

universities, public schools, government agencies, private science clubs, and amateur astronomers. The United States alone accounted for 126 groups.[10]

The endurance of Operation Moonwatch was a testament to its true popularity, as more than 5,000 volunteers equipped with telescopes and binoculars continued scrutinizing the skies over the next two decades (long after the original need had been met). More than 400,000 satellite observations were recorded. Whipple would praise the infectious enthusiasm of these early space buffs, saying, "Quite a number have gone into science or space programs because of their amateur involvement in Moonwatch."[11] After the National Aeronautics and Space Administration was formed in 1958, it would continue to fund Moonwatch at a low level through the SAO—at about $30,000 per year—until the organization was officially disbanded in July of 1975.

Despite its relatively good accuracy, optical tracking had inherent disadvantages in addition to acquisition. For instance, in order to get good photographs, the weather must be favorable and lighting must be correct. The latter condition meant that a site must be in darkness while the satellite being tracked was in sunlight. The problem with that though, is a satellite spends a significant portion of its orbit either in Earth's shadow where it is not illuminated, or in sunshine where there is not enough contrast to see it (like trying to see stars during the day). Thus optical tracking could only be used during a short window just before and after sunrise or sunset. Therefore, from any given station, it was usually impossible to get more than a few good observations of the orbit per pass.[12]

Besides optical systems, there was *radar*, an electronic system. Radar—or Radio Detection and Ranging—in which the location of an object is precisely determined by measuring returns on electromagnetic waves reflected off the object back to the transmitting station, had been a topic of research by scientists and engineers since the late 1800s when German physicist Heinrich Hertz first began experimenting with radio waves in his Frankfurt laboratory. Research continued throughout the first half of the twentieth century on both sides of the Atlantic. But it was not until World War II when the allies first used it to great advantage for early warning—in particular during the Battle of Britain—did radar come into its own. By the end of the war, it was standard operating equipment on Navy ships and aircraft on both sides.

Towards the end of the war, missile testing activities soared, and along with it, the need for reliable tracking techniques. In the fall of 1944, Major General G. M. Barnes, then Chief of Research and Development Service, U.S. Army Office of Ordnance, recognized the urgency for the Army to establish a research program in the field of guided missiles. To this end, he was convinced that as a vital part of such a program, it was necessary to have somewhere within the limits of the continental United States, a range where test firings of such missiles could be conducted routinely and safely. With this

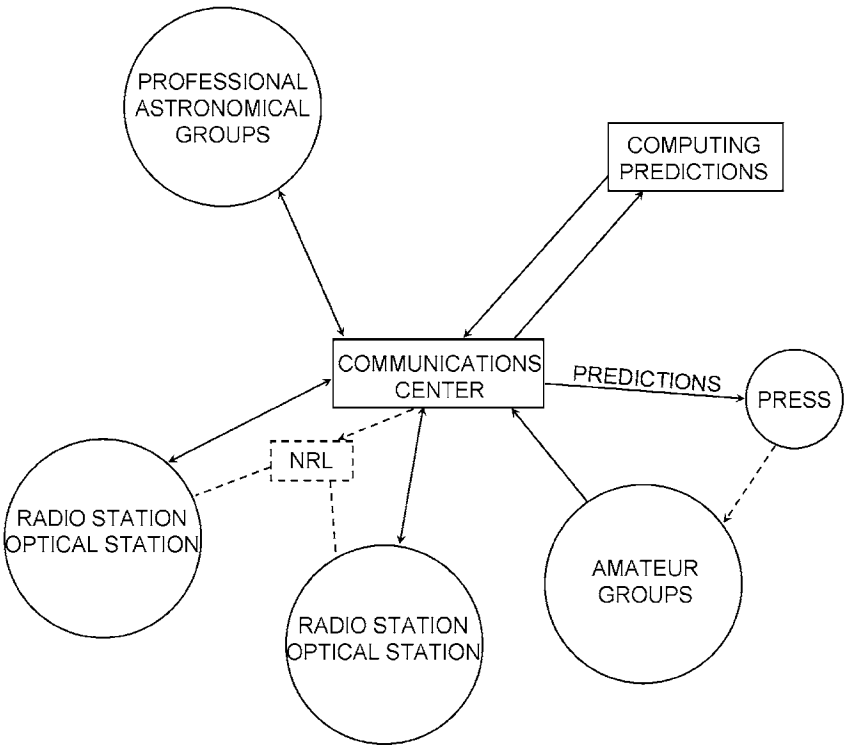


Diagram of Operation Moonwatch showing the vital role of amateur star gazers in this schematic from an original set of NRL plans. (Constance McLaughlin Green and Milton Lomask, *Vanguard: A History*. [NASA SP-4202, 1970])

in mind, military and civilian engineers from the Ordnance Department and the Corps of Engineers conducted surveys of all open areas in the United States to find a place suitable for such firing activities.

Certain fundamental requirements were set forth, such as: extraordinarily clear weather throughout the preponderance of the year; large amounts of open, uninhabited terrain over which firings could be conducted without jeopardy to civilian population; accessibility to rail and power facilities; and to whatever degree possible, proximity to communities to provide for the cultural needs of personnel to be employed at such an installation. The result of these studies indicated that a relatively desolate area around the White Sands Desert and the Alamogordo Bombing Range in southern New Mexico nearly filled all of the specifications.[13] While it was recognized that both the length and width of the range were not as large as might be desired, it could be utilized efficiently for the early types of missiles that were being developed and

for the reduced-scale prototypes of the later and larger types that the Army then had in mind. In this way, White Sands quickly became the hub of activity for U.S. missile development, and with it, tracking technology.[14]

Throughout the late 1940s and early 1950s, radar was increasingly used to supplement optical tracking. This work, at a time when the accepted practice was that optics were the primary way for data recovery, was not received with open arms by all. In spite of this initial reservation, it did have the momentum of technical advancement on its side.[15] It was at White Sands during this time that a young engineer named Ozro M. Covington began developing the concept of centralized networks for tracking and communications, honing skills which would later be used to lead NASA's human spaceflight network. After working in the field of communications and radars as a civilian in Europe and at Fort Monmouth, New Jersey during World War II, Ozzie Covington became the Technical Director of the Army Signal Missile Support Agency at White Sands in 1946 (a position he would hold until leaving for NASA in 1961.) Here, Covington and range engineers tested and refined networking communications and data processing techniques and developed what was, in essence, the forerunner to the mission control concept.

The idea of a worldwide communication complex to support spaceflight was not a new concept to many at White Sands. By the time Covington came onto the scene, it had already been the subject of study and a number of possible approaches had been considered.

> We supported the firing of V-2 rockets brought to the U.S. from Germany after World War II and, of course, our own U.S. developmental missiles, [said Covington] to monitor these flights, which could reach 100 miles (160 kilometers) in altitude and some 100 miles downrange. We established a chain of five radar tracking stations. These were linked to our 'C' Station, which also contained early computer capabilities to give us a real-time data system. We really had an early mission control center from which we not only monitored these flights but also sent guidance signals to the missiles. While we developed our electronic capabilities, we also used the real-time data to guide an array of optical instruments to keep an eye on these firings. After a V-2 went south to Juarez, Mexico, instead of north, these radars and their associated display equipment became the primary source of data for the range safety officer.[16]

Among these first networks was the very successful tracking system developed at the range using an elaborate chain of FPS-16 C-band, single-object tracking radars made by the Radio Corporation of America (RCA) to track missiles launched on extended horizontal flights. The units

were networked to transmit data to a control center where consoles displayed information on the radar returns for the test engineers. Operating either on commercial line power or generators, these transportable radars could be moved from one place to another or to undeveloped sites using prefabricated transportable pads so as to provide support from different locations all around the range. A modernized version of this radar tracking system continues to be used at White Sands today.[17]

Still, a third approach for tracking emerged in the late 1940s: *radio interferometry*. Like most electronic tracking techniques then in use, radio interferometry required the presence of a signal source, or transmitter beacon, on the object being tracked. Despite its somewhat intimidating name, the basic operating principle of radio interferometry is relatively straight forward. Since radio signals—like sound—travel in waves, separate antennas will receive the waves at slightly different times. Knowing that electromagnetic waves propagate at the speed of light, measurements of the different arrival times of the waves, called phase differences, can then be used to calculate position solutions for an object. This technique had the advantage of yielding highly accurate tracking angles and could be used under virtually any atmospheric condition, even underwater. Given its potential, a number of groups were involved in its research after World War II. A significant milestone was reached in 1948 when engineers of Consolidated Vultee Aircraft Corporation (Convair) created the first working application for the U.S. Army on the Azusa missile tracking system.

To understand how this technique was utilized and eventually developed into what would become the Minitrack Network—the progenitor to the STDN—one has to go back to the year 1955. That summer, plans were drawn up by the United States to launch the first artificial satellite as President Dwight D. Eisenhower announced on 29 July that the United States would participate in the upcoming IGY by orbiting a 9-kilogram (20-pound) satellite. In keeping with the spirit of the IGY, Eisenhower announced that the American satellite program would be civilian in nature. Despite the announcement, it was well understood by the science community that assistance from the military was going to be required, since at the time, it alone possessed the capability to launch a rocket large enough to orbit a satellite.

Following this announcement, a Committee on Special Capabilities was appointed by the DOD to select the appropriate organization to lead this effort. This committee would end up selecting a proposal submitted by the Naval Research Laboratory (NRL) as the one which seemed best qualified to promote the civilian nature of the project while at the same time support military interests in space technology development. The NRL was a good choice. It was well qualified, having already conducted research in this field for nearly a decade, even developing an underwater tracking system using sound inter-

ferometry. By the early 1950s, the NRL detachment at White Sands had built and fielded a rudimentary tracking system for the development of the Viking short range guided missile.

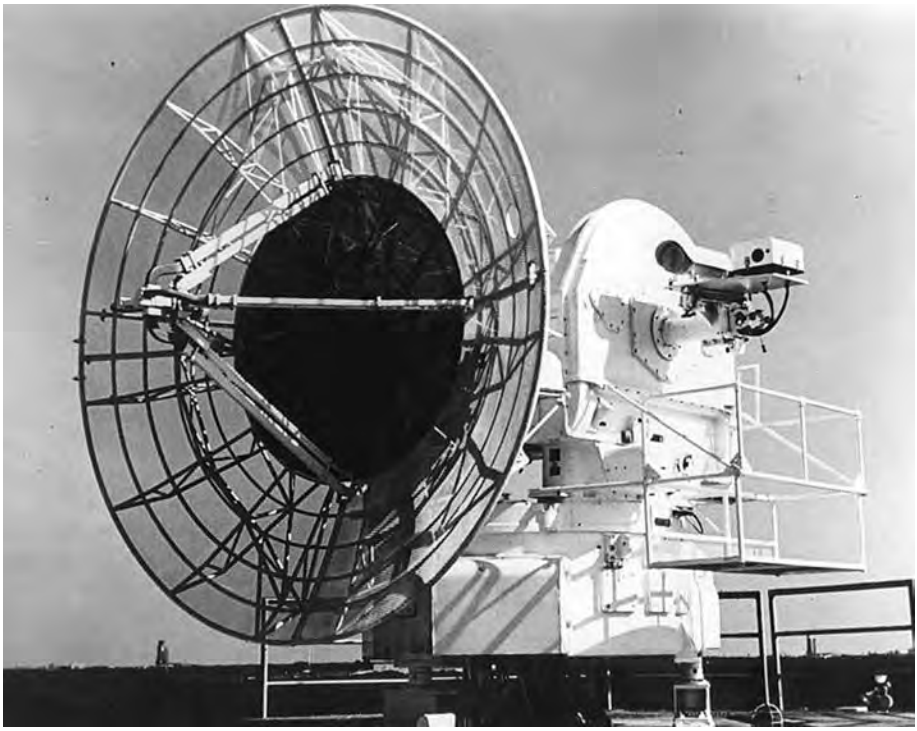
In this way, the effort selected by the committee, called Project Vanguard, thus became the United States' official IGY entrant in September 1955. The project was led by Milton W. Rosen, Chairman of the NAS Advisory Committee on the IGY, and was managed jointly by the Office of Naval Research and the NAS Committee on the IGY.[18]

Once Vanguard was given the go-ahead, a significant portion of the early work was devoted to solving the problem of just how to track such a small object in space, since the Vanguard satellite would only be 15 centimeters (6 inches) in diameter. Most atmospheric research scientists were advising Rosen's team to rely on optical tracking; it was, after all, the tried and proven method. In addition, camera observations of meteoroids entering Earth's atmosphere had demonstrated that modern terrestrial optical instruments could indeed spot small objects weighing only a few pounds that moved at high velocities in the upper atmosphere.

Rosen was well aware of the limitations of optical tracking—in particular, the acquisition problem—and wanted something better. In other words, optical instruments could see a satellite, but could they find it? To get a better handle on the problem, he had Richard Tousey at the NRL verify visibility computations that had been performed earlier. Tousey's answer confirmed what Rosen had already suspected. In his opinion, the probability of successful optical acquisition of a Vanguard-sized satellite on the first visual pass was only $1x10^{-6}$, that is, literally one in a million.[19]

Thus, convinced from the beginning that his team must look elsewhere for a solution, Rosen asked John T. Mengel and his NRL Tracking and Guidance Branch to develop an electronic system for use in conjunction with an optical one. For ballistic missile guidance, Mengel had led the development of the mentioned Azusa system at White Sands. For tracking a satellite, however, that system was out of the question as it required an onboard transmitter weighing 13.6 kilograms (30 pounds), far too large for a small satellite. Mengel's team immediately went to work and within a year came up with an interferometer system that, although based on the techniques developed for Viking, required only a 0.37-kilogram (13-ounce) transmitter using a different operating frequency and antenna pattern. It was, in essence, a pioneering new system.[20]

Hired by nuclear submarine pioneer Hyman G. Rickover a year after the Japanese attack on Pearl Harbor, Jack Mengel, who later became the first Director for Tracking and Data Systems at the Goddard Space Flight Center, began his career for the NRL working on antisubmarine devices. "At first, I was responsible for telemetry and radio control systems in the Rocket



The RCA FPS-16 radar has been widely used at missile ranges since the 1950s. It is a transportable, Single-Object Tracking Radar which operates in the C-band of the electromagnetic spectrum. The photograph shows a FPS-16 at Patrick Air Force Base used in 1961 to track launches out of Cape Canaveral. (Photo courtesy of Patrick Air Force Base Office of History)

[Sound] Branch, which used V-2 rockets of World War II fame for scientific observations in the near-Earth regions. This was followed in 1950 by work on a Viking missile project and its possible use in submarine warfare."[21] Here, his work in developing an X-band interferometry system that operated at the centimeter (one inch) wavelength provided the high precision missile guidance that the U.S. military was looking for.

Expounding on this initial work, he and his team then converted this system into a 108 megahertz (MHz) system for tracking satellites. This new radio interferometry satellite tracking system had the catchy name Minitrack, which Mengel derived from "minimum-sized tracking system." The name was fitting, as tracking was accomplished using a small oscillator onboard the satellite that illuminated pairs of antennas at a ground station with which angular positions of the satellite was derived using the radio inter-

ferometry technique. "What apparently had sold the Defense Department on our proposal was the fact that it consisted of a good radio system which would be infallible in picking up the satellite in orbit," explained Mengel.[22] This pioneering tracking network was to become operational in 1957 at a cost of only $13 million.[23]

Despite the low price tag, America's first spacecraft tracking network had to overcome several key technical hurdles in order to achieve a level of tracking accuracy never before attempted. Angle measurements and accurate timing were vital. One had to know precisely at any given instant in time where a satellite was in orbit. From the very beginning, the ability to ascertain precise timing, in particular, was crucial. Chesley H. Looney, Jr., the former Head of the Time Measurement Branch for Minitrack at the GSFC said:

> In the early days of the space program, prior to the establishment of NASA, we had to develop a reliable, low-frequency phase-measurement system to track our Vanguard satellites. While a high frequency interferometer system had been used at White Sands Missile Range, New Mexico, to track military rockets, this system was inadequate to meet the needs of low-powered satellites then being planned. Thus, a new system at a different frequency and with a different antenna layout to provide the proper precision and signal resolution had to be designed . . . In those days, our budgets were very tight and we were forced to use low cost hardware. Yet, the equipment had to be capable of phase-angle measurements with an accuracy not attempted before. In the process, we developed an elegantly simple, filter-inductor system. In our related time measurements, we achieved an accuracy of 10-milliseconds per day and broke new ground in phase-angle tracking later embodied in commercial designs.[24]

As eventually implemented, Minitrack used a 10-milliwatt, 370-gram quartz crystal controlled oscillator transponder aboard the satellite which operated at the fixed frequency of 108 MHz. It had a battery life of 10 to 14 days, sufficient for a planned Vanguard mission. The new tracking network was considered quite revolutionary for its time and was thus subject to much scrutiny by the general scientific community. In a series of papers and speeches over the next two years, Mengel and his colleagues—notably Roger L. Easton, his assistant at NRL, and Paul Herget, Director of the Cincinnati Observatory and a consultant on Project Vanguard—responded to this scrutiny by presenting the difficulty of tracking and its importance in the U.S. satellite program.

In a March 1956 paper, Mengel explained the challenge of tracking an artificial satellite to fellow scientists and engineers. In it, he said:

> The final realization of man's efforts to place a satellite in orbit about the Earth will immediately pose a new series of problems: How to determine the precise orbit that it is following and how to measure what is happening within the satellite from the vantage point of a ground station. The immensity of the first of these programs, how to prove that the satellite is in fact orbiting—the acquisition phase—can be realized by an analogy. Let a jet plane pass overhead at 60,000 feet (18,000 meters) at the speed of sound, let the pilot eject a golf ball, and now let the plane vanish. The apparent size and speed of this golf ball will approximate closely the size and speed of a satellite three feet in diameter at a height of 3,000 miles (4,800 kilometers). The acquisition problem is to locate the object under these conditions, and the tracking problem is to measure its angular position and angular rate with sufficient accuracy to alert nonacquiring tracking stations, those trying to follow the satellite by optical means, as to the time and position of expecting passage of the object.[25]

In describing how Minitrack worked, Mengel and Herget likened the antennas of a station to human ears. In short, it used stereo-phonics to pinpoint the location of a given signal source. For instance, a person locates the source of a sound by virtue of the phase difference in the sound waves which arrive at slightly different times at each of his ears. The antennas of a Minitrack station functioned much like ears. The electronic receivers of the Minitrack system were connected to pairs of receiving antennas set at a known distance apart, which indicated the direction of the signal by phase differences in the radio wave that it received from the satellite. The direction to the satellite, measured relative to the baseline vector between the two antennas, could then be calculated from the phase difference using triangulation.

For example, if the waves arriving at two antennas were out of phase by one-third of a wavelength, the extra distance they traveled to the farther antenna could be deduced and the angle to the baseline calculated using trigonometry. However, in order to find the actual length (that is, magnitude) of the extra distance, it must be known whether this phase difference was only one-third of a wavelength or some multiple thereof. To resolve this ambiguity, other pairs of antennas were set up but spaced closer together at distances of less than one wavelength. These arrays provided shorter fractional phase differences, and on the basis of whether the resulting directions fell within the reception pattern of the antenna system, the number of wavelengths involved could be determined.

A Minitrack antenna array was quite large, consisting of a 150-meter (500-foot) cross of eight linear dipole antennas (metal frame antennas

with many elements like the traditional rooftop, television antenna) mounted above 18-by 3-meter (60-foot by 10-foot) ground screens which served to eliminate multipathing, or interference, from ground reflection. In the north-south direction, three pairs of antennas spaced 150, 20 and 4 meters (500, 64 and 12 feet) apart, respectively, were used; in the east-west direction, two pairs with spacings of 150 and 20 meters were used. The resulting reception pattern was a fan-shaped beam spanning 100° north-south by 10° east-west, measured at the 6 decibel (dB) reception power point, a common demarcation used to measure antenna reception strength.[26]

In this setup, the north-south and east-west determinations provided the two angles needed to locate the actual direction to a satellite. However, more information was still needed. To verify precise antenna alignment (that is, its orientation with respect to a fixed celestial coordinate system) and timing, calibrations had to be done. This was done periodically by having an airplane equipped with a Minitrack transmitter and flashing lights fly over the station at night at an altitude of 6,100 meters (20,000 feet). An optical system called MOTS—Minitrack Optical Tracking System—was used for this purpose.

MOTS was one of the lesser known tracking systems implemented early in the network. Developed under the direction of Edmund J. Habib at the GSFC, stations used it to calibrate and determine position misalignments in the Mintrack system. It worked something like this: A time-coded lamp on the belly of the airplane, operating simultaneously with an onboard 108-MHz Minitrack transmitter, allowed for precise observation of the airplane's position against the background star-field as the Minitrack array on the ground recorded the movement of the Minitrack beam. In this way, a precise comparison between the optical position (aircraft belly light) and the radio frequency position (aircraft transmitter) could be made for calibration purposes.[27] Precise optical and radio fixes could then be made to correct for any misalignment of the antenna. Using this setup, Minitrack was sophisticated enough to record phase differences to an accuracy of one-thousandth of a wavelength and the time of observation to one millisecond, which in turn produced a direction fix to the satellite accurate to within 20 arc-seconds (1/180th of a degree). Such was the kind of resolution that was needed in order to track Vanguard.[28]

As implemented, the Minitrack stations were all very similar. Their principal antennas consisted of the large arrays for angular tracking, one fixed antenna array for telemetry reception, and a single rhombus (diamond shaped) antenna for communications. Three trailers housed the ground station core electronics consisting of telemetry recording equipment and communications hardware. Site selection had to consider many factors. Due to the size of the main array, a station required a large plot of land, at least 23 acres in an area that was relatively smooth, with a gradient of less than 1°. To maximize the observation time during each pass, Minitrack specifications called for the ter-

rain adjacent to a station to not exceed an elevation angle of 10° for at least 800 meters (one-half mile), and 20° for 8,000 meters (5 miles). Finally, to minimize interference, engineers selected locations away from electric power plants and even airports (although this recommendation was not followed at many of the sites eventually constructed).[29]

The final decision by the IGY selection committee on the U.S. satellite project came down to two proposals: the Navy's Project Vanguard and a competing concept called Project Orbiter submitted by Army scientists—including Wernher von Braun. The latter, however, originally contained no provision for electronic tracking. While it may be too simple to say that the NRL proposal was selected because it had provisions for electronic tracking, it is clear that the inclusion of a Minitrack Network in the NRL proposal undoubtedly had a direct bearing on the committee's decision in August 1955. Mengel later supported this claim, saying, "What apparently sold the Defense Department on our proposal was the fact that it consisted of a good radio system which would be infallible in picking up the satellite in orbit."[30]

In addition, NRL also described in their proposal a somewhat less elaborate version of the Minitrack system as eventually implemented. Developed by Roger Easton, this toned-down system became known as the "Mark II," or unofficially, the "Jiffy." Some in the halls of NRL referred to it as the "Poor Man's Minitrack." Using less-sophisticated radio frequency phase-comparison (instead of interference), the Mark II, nevertheless, went on to become the nucleus of Project Moonbeam, a program sponsored by NRL to encourage amateur radio enthusiasts and their clubs to build their own ground stations to track satellites.[31]

With its selection accomplished, design and construction-preparation of the Minitrack Network began in the fall of 1955. From its inception, Minitrack was a project that had to have cooperation amongst all the stakeholders. Considering all the technical and administrative requirements that were needed to coordinate the work of various military units, university laboratories, individual experts and private contractors, its progress was remarkably smooth. Years later Mengel was able to recall "a few personality clashes" but in his opinion, those were "par for the course for a program that made use of some of the best astronomers in the country."[32]

During initial development and construction, the U.S. military played the decisive role. Three Army agencies—Corps of Engineers, Map Service and the Signal Corps—were responsible for most of the actual construction. The Air Force's main role was the installation of tracking radars at Patrick Air Force Base in Florida and on Grand Bahamas Island. Overall program management was under the Naval Research Laboratory. In addition, the Navy's Bureau of Yards and Docks obtained use of the land needed to construct the prototype station at Blossom Point, Maryland. Outside of the military, the Department of



Minitrack antenna arrays were large, spread over an area one and a half times the size of a football field. (Constance McLaughlin Green and Milton Lomask, *Vanguard: A History*. [NASA SP-4202, 1970 file 6589 NASA Historical Reference Collection, NASA History Division, Washington, DC 20546])

State provided the much-needed assistance to the Navy as the executing agent for foreign affairs, conducting negotiations for land leases on foreign soil. This cooperation between the various departments of the U.S. government was a testament to the priority of Project Vanguard. There was with it a definite sense of urgency. Chesley Looney recalled years later that, "While our budget was low, . . . [we] had high priority which allowed us to push ahead without administration hurdles. All of us were anxious to succeed."[33]

Things moved quickly. In the spring of 1956, a project advance team led by Commander Wilfred E. Berg, surveyed 17 Latin American sites from which six were selected. They were:

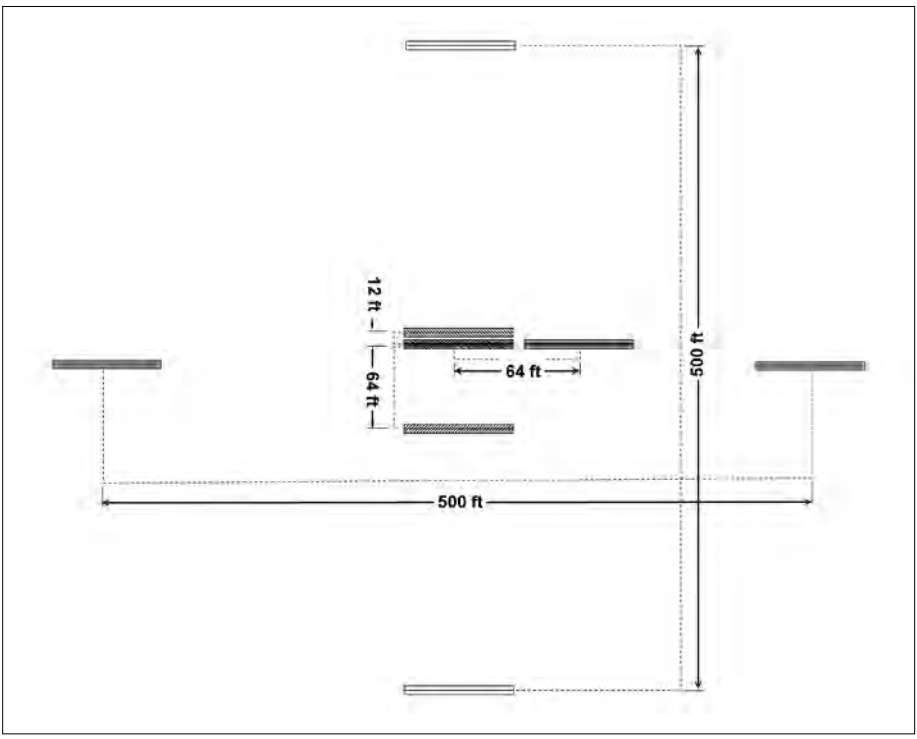
Batista Field in Havana, Cuba
Paramo de Cotopaxi near Quito, Ecuador
Pampa de Ancon at Lima, Peru
Salar del Carmen at Antofagasta, Chile
Peldehue Military Reservation near Santiago, Chile
Rio Hata in the Republic of Panama.



One of the lesser-known tracking systems developed by GFSC was the MOTS—Minitrack Optical Tracking System—which was used to obtain precise, position calibration and misalignments in the Minitrack system. The photograph shows David W. Harris (who later became a Deputy Associate Administrator at NASA Headquarters) operating the camera at Blossom Point in 1963 after it was modified to photograph passive satellites such as Echo. Note the photographic plate at the bottom of the camera. MOTS was used until the 1970s. (NASA Image Number G-63-826)

Shortly after Berg's group returned in early May, NRL eliminated the Panamanian site based on studies indicating that a station at San Diego would be more useful for early orbit determination. This decision would make the station on the coast of California the first to receive confirmation from Vanguard that it indeed had successfully attained orbit.[34]

As the network began to take shape, there emerged basically three categories of stations. The first type involved three tracking units located immediately downrange of Cape Canaveral on the islands of Grand Bahama, Antigua, and Grand Turk, networked with a radar installation at Patrick Air Force Base, Florida. These monitored the launch, ascent, and orbital inser-



The antenna system of a Minitrack station was made up of eight individual antennas connected to form three pairs in the north-south direction and two pairs in the east-west. The outermost pairs in each direction were 152 meters (500 feet) apart while the inner north-sourth pairs were separated by distances of 19.5 and 3.7 meters (64 and 12 feet). The spacing of the inner east-west pair was 19.5 meters. (Mengel and Herget, Tracking Satellites by Radio. Folder 8800, NASA Historical Reference Collection, NASA History Division, NASA Headquarters, Washington DC)

tion ("early ops") of the Vanguard vehicle. In the second category were the so-called primary stations located roughly on a north-south line along the east coast of North America and the west coast of South America so as to form a "picket line" across the expected path of satellites. To preclude over-flight of the Bahamas, launches out of the Cape were limited to orbital inclinations of 35° or less (in other words, the ground track of satellites were bound between 35° in latitude north and south of the Equator). With this picket line, radio signals from a satellite could be intercepted as it crossed the 75th West meridian on each orbit. Seven primary stations formed this line:

Blossom Point, Maryland
Fort Stewart, Georgia
Antofagasta, Chile
Santiago, Chile
Quito, Ecuador
Lima, Peru
Havana, Cuba[35]

Lastly, there were other stations outside of the picket line in locations around the world at San Diego, California; Hartebeestpoort outside of Johannesburg, South Africa; and Woomera, South Australia. This last group of stations was needed to help determine the initial orbit of the satellite. With this arrangement, Mengel estimated that there was a 90 percent chance of capturing every pass of a satellite which was at an altitude of 500 kilometers (300 miles) or higher.

The first station established was on a 23 acre NRL managed site in the U.S. Army Blossom Point Proving Ground, Maryland, some 60 kilometers (40 miles) south of Washington, DC. The Army had specifically permitted this portion of the range to the Navy in 1956 for use as a Vanguard support site. Bordering the Potomac River, Blossom Point provided an ideal setting for a ground station, with horizon-to-horizon look angles along with an interference-free, low-noise environment. As the network developed, this prototype station functioned somewhat like a test station for the rest of the network. Here, engineers, scientists, and station operators received their training, and procedures were developed for system tests, simulations, and checkout, including use of the MOTS cameras.

When it was first conceived, NRL engineers envisioned using only four sites, along with the test-bed at Blossom Point. By the time Vanguard was ready to issue its first full progress report in December of 1955, however, Mengel's team was thinking in far more elaborate terms. By the time it was completed, the Minitrack Network would consist of 14 sites, including the DOD launch support radar stations. With the obvious need to place ground

stations in other countries, the ability to secure foreign cooperation became paramount. In fact, the ability to work effectively with local governments and foreign nationals would play directly into how well stations could accomplish their missions in the years to come. This NASA hallmark began in the days of Minitrack. As B. Harry McKeehan, Chief of International Operations at the GSFC from 1963 to 1980, explained:

> Very early in our planning for the tracking network, there emerged a definite commitment that our foreign tracking stations were to be conceived carefully and operated with the full support and active cooperation of the foreign governments involved. Thus, even in times of political turmoil, we were usually able to continue our operations without serious interference. . . . [In] instances when stations had to be closed because of political conditions within the country, even in those cases, we continued to recognize and honor our commitments to the host government and our local employees.[36]

McKeehan soon found that one of the best ways to secure international cooperation was to involve local nationals in station operations. "A major stabilizing factor in our foreign operations was our policy of maximum utilization of local people in station positions. This proved to be cost effective as well as giving the local government and population a feeling of having a stake in the NASA missions." McKeehan explained how this "nationalizing of the stations" was done.

> Our efforts normally started after the technical requirements for a specific program had been identified and the tracking coverage determined. Then, we explored with our colleagues in NASA Headquarters and the appropriate officials of the Department of State, the country to be approached for a 'tracking partnership'. We considered the country's geographic features, accessibility, political stability and available logistic support. Also, we looked into possible sources for local employees to reduce the number of U.S. personnel. Then, we would work through the U.S. embassy, establishing contacts with representatives of the foreign governments and explaining our needs to their leaders. This approach, together with the worldwide excitement of the U.S. space effort, opened doors and usually assured us of the support needed.[37]

To meet technical standards, NASA provided training and other educational programs for local employees which ultimately ended up also

broadening the scientific base in those countries. "With concern for the sensitivities of other countries, we learned to manage responsibly the resources and manpower needed to assure our success in exploring space. It helped create a great reservoir of goodwill throughout the world for the U.S. space effort and . . . proved to be a valuable future investment," said McKeehan.[38] Even today, as the space agency continues to promote international goodwill—albeit in a very different global environment—the lasting effect of this policy can still be felt, the roots of which date back to this time.

As site construction finished, all three military branches also contributed to their operation and maintenance. Army personnel operated the five prime Minitrack stations in Latin America plus the one at Fort Stewart, Georgia. Meanwhile, the Naval Electronic Laboratory operated the station at San Diego, Blossom Point, and the tracking units on Grand Bahama Island, Antigua, and Grand Turk. The Air Force operated their radar installations at Patrick and Grand Bahama. In Australia, the method of operation was somewhat different. There, the Weapons Research Establishment (WRE), an agency of that commonwealth's Department of Supply (DOS), constructed and operated the station just outside of the village of Woomera. Under agreement with the Australian government, the DOS supplied the land, power, facilities, and workers. In return, the United States furnished all the technical equipment, trained the WRE personnel, and installed the initial equipment. Half way around the world, an agreement was reached with the South African government allowing for the National Telecommunications Research Center to staff the station just outside of Johannesburg with South African nationals.

Finally, the NRL contracted with Bendix Radio in Towson, Maryland, a Division of Bendix Aviation Corporation, to build the station electronics. Being the equipment supplier for the DOD, Bendix was a logical choice to provide this service. (After NASA was formed and operation of Minitrack was turned over by the DOD to the new space agency, Bendix would quickly assume operations of the stations in addition to just providing the hardware.)

One of the early challenges of Minitrack was the need for rapid data processing in order to compute and determine a satellite's orbit. To this end, NRL began working to implement a solution very early on in the network design process. As early as September 1955, Paul Herget met with Cuthbert C. Hurd, then Director of Electronics Data Processing Machines, and other officials of International Business Machines (IBM), to discuss the computational requirements of the Earth satellite program. In March of the following year, the Office of Naval Research issued a request for proposals for the renting of computer facilities and the furnishing of mathematical and programming services for Project Vanguard. IBM and two other companies responded with IBM winning the contract at a bid of $900,000.[39]

Under the contract, IBM was to supply personnel to operate, around-the-clock, its 704 computer for six weeks after installation and checkout, and then to provide, free of charge, orbital computations over the life of the first three successful satellites or lifetime of the Minitrack Network—whichever occurred first. In July 1956, NRL's new Vanguard IBM Computing Center opened in downtown Washington, DC. John P. Hagen, Director of Project Vanguard, appointed a committee led by Joseph W. Siry, Head of the Theory and Analysis Branch at NRL, to oversee the orbital computation work. Joining Siry on the committee were Gerald M. Clemence and R. L. Duncome of the Naval Observatory, and Paul Herget from the University of Cincinnati.[40] Since this was a new center, Siry had IBM network the facility via a transceiver to a backup site located at the company's Research Computing Center in Poughkeepsie, New York. This ensured computational redundancy and reliability that the NRL felt it needed.[41]

Network operations could be summarized as follows: Within minutes after a satellite pass, observations from a station were transmitted by teletype to the Vanguard Control Center. After being (manually) inspected for errors, the data was recorded onto reel-to-reel magnetic tapes where they were taken by courier to the Computing Center. There it was transferred onto punched cards and fed into an IBM-704 analog computer (better known at the time as an "electronic calculator"). The machine was programmed with mathematical calibration formulas and correction factors for each station to account for certain systematic errors such as atmospheric conditions and refraction of the radio signal as it passed through the ionosphere. Distances and directions from the given ground station to the satellite were then computed.

As observations on various points along the trajectory were collected, the calibration formulas were updated and the satellite path calculated (a process known as differential correction and orbit determination). Adding higher order corrections to account for parameters such as atmospheric drag and nonuniformity of the Earth gravitational model—for example its nonspherical "pear shape"—and variations in geographical mass concentrations, the IBM-704 was able to compute (and predict) a satellite's position and velocity 150 times faster than the actual progress of the vehicle. This was certainly no small feat in an era before the digital computer. In this manner, the orbit of the satellite was steadily refined as it made more and more revolutions around Earth.[42]

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By September 1957, Minitrack was in place. To the disbelief of its builders, however, the network would begin its operations by tracking not an American satellite but a Russian one. Just three days after completing a checkout of the network on 4 October 1957, the Soviet Union shocked the world

by announcing the launch of Sputnik 1 (the name meant "satellite" or "fellow traveler"). Once they were sure that their 45-kilogram, beach ball-sized sphere had indeed made it into orbit, the Soviet news agency Tass announced from Moscow that radio signals from the satellite could be heard by receivers set at 20 and 40 MHz. Amateur radio operators from all over the world scrambled to pick up the steady beep of the transmission.

The unexpected Soviet announcement disrupted Minitrack preparations as all the equipment had been set up to operate at the planned American satellite frequency of 108 MHz. In the words of one space pioneer, the United States was "caught with its antennas down."[43] Stations had to quickly convert their Minitrack receivers to operate on the new frequency and to put as many stations as possible into operation before Sputnik's transmitter gave out. Unlike today where receiver frequencies can be changed by the mere turn of a knob, entire racks of equipment had to be changed out. Instructions sent by teletype and equipment flown in by the Army enabled stations at Blossom Point, San Diego, Antofagasta and Lima (and later at Santiago and Woomera) to convert to operating at 40 MHz within a few days.

Although the Central Intelligence Agency (CIA) had been fully aware that the USSR was planning to put something into space, the actual launch nevertheless still caught American space planners off guard. Chesley Looney illustrated just how much so:

> When the Soviet Union launched its Sputnik in 1957, we were totally demoralized initially. This lasted for about 24 hours. Then we pulled up our socks and quickly put together a 40 megahertz system and three boxes of antenna gear which we flew to Lima, Peru, and to Antofagasta and Santiago, Chile, and quickly started to track this new bird. Unfortunately, it was not ours, but while we were not happy, we gained some valuable experience for the first Vanguard vehicle tests expected to follow in December 1957.[44]

Before its batteries died some 14 days later, the Naval Research Laboratory was able to collect some very useful tracking information on Sputnik 1 (and later Sputnik 2). Despite the obvious American disappointment, the whole experience gained tracking the first two Soviet satellites ended up being very useful for Minitrack, in essence providing a "real world" opportunity to check out communication lines, time references, clocks and data transmission accuracy. The United States had chosen a frequency of 108 MHz because it would give a more accurate indication of direction than the lower frequencies. For example, a radio signal transmitted on a carrier frequency of 40 MHz is bent approximately seven times more sharply than one at 108 MHz; for 20 MHz, it is 29 times. Therefore, a scientific fallout from

tracking Sputnik was the U.S. realizing a large amount of data from which new knowledge about the electrified layer of the ionosphere and its effect on radio wave propagation was gained.

Two months after the stunning blow of Sputnik 1, the United States was ready to attempt orbiting its own satellite. On 6 December 1957, the Navy's Vanguard rocket, with a tiny 1.5-kilogram (3.3-pound), grapefruit-sized satellite inside the nosecone, rose a few feet off its launch pad at Cape Canaveral, and then to the dismay of those watching, promptly crashed back down on itself erupting into a huge ball of fire. Unlike the Soviet Union's delayed announcement of Sputnik, the launch attempt of Vanguard was seen on American television. The psychological impact of this spectacular failure could not be understated. Just one month earlier, the Soviets had succeeded in orbiting a dog on the 450-kilogram (990-pound) Sputnik 2. Five days after the failed attempt, President Eisenhower, who up until then had been publicly endorsing the Navy's Vanguard as the official U.S. satellite program, reluctantly (and quietly) granted the Jet Propulsion Laboratory under the direction of the Army the approval to attempt a satellite launch.

Three months later on 31 January 1958, the Army came through. Explorer 1 was launched into orbit atop a Jupiter C IRBM to become America's first artificial satellite. In addition to a 108 MHz transmitter (which operated for a remarkable 113 days), the 68-kilogram (31-pound) satellite had onboard an experiment devised by Professor James A. Van Allen of the University of Iowa to detect charged particles in Earth's magnetic field. In time, telemetry downlinked from the first four Explorer satellites led to the discovery of the Van Allen Radiation Belts, hailed as the single most important discovery of the IGY.

Seven weeks later on 17 March, the Navy—on its third attempt—finally followed Explorer 1 into space with the launch of Vanguard 1. Although its success was delayed, the satellite and its all-important accompanying tracking network were to pay great dividend almost immediately. Utilizing the precise tracking capability afforded by Minitrack, telemetry provided by this tiny sphere, which when analyzed, showed Earth's geode to be slightly pear-shaped rather than a perfect sphere—a fact verified some 10 years later in beautiful color photographs taken by Apollo astronauts on their way to the Moon.

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Spurred on by the sudden urgency brought about by Sputnik, President Eisenhower in early 1958 revisited an initiative that up until then had not exactly occupied high priority on his administration's agenda. That is, a national governing agency whose charter was to develop, direct and coordinate all American space-related activities.

NASA, as it turned out, was not to be a completely new creation, but was instead, transformed from a predecessor organization, the National Advisory Committee for Aeronautics, or NACA. Founded in 1915, NACA was a civilian U.S. government agency whose original charter was to advise on and coordinate research that was being conducted by other organizations in the (then) new field of aeronautics. However, it quickly established itself as the leading research organization in the field, pushing back the envelope of flight, from supersonic flight in 1947 to high altitude experimental aviation that would set the foundation for eventual human flight into space.

After Sputnik—throughout the remainder of 1957 and into 1958—the status and progress of existing U.S. space activities was reexamined by Congress, Eisenhower, and the military. The nature, scope, and organization of long-range space goals were debated. The principal issue at hand was not *whether* there should be an organized space program but rather *the extent of military involvement*. To settle the issue, Congress conducted several hearings by the Military Preparedness Subcommittee of the Senate Committee on Armed Services from November to January. Eisenhower himself advocated civilian control of existing space related activities but showed little progress in resolving the broader, long range issues. However, Congress soon forced the administration's hand.

In January 1958, it introduced a bill authored by New Mexico Senator Clinton D. Anderson on behalf of the Joint Committee on Atomic Energy. The bill amended the Atomic Energy Act by giving the Atomic Energy Commission a major share of the nation's space program. Not surprisingly, the Administration decidedly opposed this bill. But it was put in a position where it had to propose an alternative. On 4 February 1958, President Eisenhower announced that he was assigning James R. Killian, Jr., the responsibility of developing a definitive solution. (Killian, who was at the time President of the Massachusetts Institute of Technology, took a leave of absence between November 1957 and July 1959 to serve as the first presidential science advisor.) Following the Sputnik crisis, Eisenhower established the President's Science Advisory Committee (PSAC) of which he appointed Killian chairman.

A month later, on 5 March 1958, President Eisenhower received the answer he had wanted, approving the recommendations of the PSAC that the "leadership of the civil space effort be lodged in a strengthened and redesignated National Advisory Committee for Aeronautics and that legislation be enacted to give NACA the authority and flexibility" to carry out its expanded responsibilities.[45] Politics aside, the recommendation was not unexpected. NACA had begun posturing itself in January of that year by proposing an interagency space program made up of leadership from its own organization, the DOD, the National Science Foundation (NSF) and the NAS. Meanwhile,

NACA's leaders—in particular Hugh L. Dryden—had close ties to many on the President's advisory committee, including Killian himself and World War II aviation hero, General James Doolittle.

Killian called the President's 5 March endorsement the conclusive act with respect to building a civilian space agency upon the NACA structure. Several factors favored into this decision: 1) NACA was already an established research Agency with a large technical staff and sizable research facilities; 2) NACA had been progressively moving into space research; 3) If NACA would not be allowed to move further into the space field, its whole future would be in jeopardy; 4) NACA had a long history of close cooperation with the DOD; and 5) NACA's liabilities could be overcome.

This last point was important as the committee specifically identified what these liabilities were: 1) NACA did not have an across-the-board space competence nor did it have much experience in the administration of large-scale developmental contracts; 2) Most of the nation's space work had been done by or for the DOD, and NACA would have had to tap this competence without impairing the military space program; 3) NACA was not in a position to push ahead with the immediate demonstration projects which may be necessary to protect the nation's world prestige; 4) NACA was limited by the somewhat inflexible hiring and pay provisions of civil service regulations; and 5) NACA's organization and procedures were geared to a much lower level of expenditure than would be the case after its expansion. Nevertheless, the advisory committee concluded that these liabilities could be overcome—or at least be mitigated—by enacting appropriate legislation.

Towards this end, four specific recommendations were proposed. First, NACA should be renamed the National Aeronautics and Space Administration. Second, it should be permitted to establish pay rates in excess of those of the Classification Act of 1949. Third, the Agency Head should be appointed by the President with the advice and consent of the Senate.[46] Finally, the composition of the 17-member NACA governing committee should be changed.[47]

By far the most important point to come out of the 5 March 1958 Presidential memorandum was the clear emphasis on the civilian nature that the new U.S. space agency should have. Language to the effect permeated the memorandum, such as "entail increased expenditures and the employment of important numbers of scientists, engineers, and technicians," and that "an aggressive space program will produce important civilian gains in general scientific knowledge and the protection of the international prestige of the United States," and straight to the point that the "long-term organization for the federal space program . . . should be under civilian control."[48]

In making the recommendation, the PSAC cited several other options which were considered but in the end did not have the advan-

tages that transforming NACA would have. For example, the DOD was not recommended because its deep involvement in the ballistic missile program conflicted with the desired civilian scientific emphasis. The Atomic Energy Commission was not recommended because its charter was not the space program. Finally, putting the civil space program under a proposed Department of Science and Technology was not recommended due to the delays and the process that would have been required with creating a brand new government department, something that the Administration did not want to have to go through.

On 2 April, one month after President Eisenhower approved the plan, the White House set the process in motion. A draft legislation establishing NASA was sent to Congress by the Bureau of the Budget, and a White House directive was issued to NACA and the DOD instructing them to take certain preparatory actions pending congressional approval on the draft legislation. This draft to Congress was accompanied by a personal message from Eisenhower to congressional members urging them to enact appropriate legislation in order to promptly establish a national space program which he deemed was essential to the general welfare and security of the nation. He added that the space program should be given high priority and, of course, be soundly organized.

In summarizing its civilian nature, the President said: "I recommend that aeronautical and space science activities sponsored by the United States be conducted under the direction of a civilian Agency, except for those projects primarily associated with military requirements. . . . The civilian Agency should be a new one and include aeronautical activities as well."[49] In anticipation of congressional approval and to pave the way for the new Agency, the White House initiated a series of preparatory actions under the supervision of Dr. Killian and the Bureau of the Budget.

In May 1958, it instructed NACA to prepare a full explanation of the proposed legislation for presentation at congressional hearings. NACA was to also make plans to reorient its programs. Internal organizations and management structure were changed to carry out the functions to be assigned to the newly formed National Aeronautics and Space Administration, and as such, to lay the groundwork for whatever expansion might be necessary in order to implement the legislation when enacted. Along these lines, the White House directed the DOD and NACA to jointly review current Defense space programs to determine which ones should be transferred to the new Agency, and of those transferred, which would require continued military support. Finally, Killian wanted NACA officials to meet with the NAS, the NSF and the country's academic community so as to ascertain how one might best secure their participation at large.

By the time the draft legislation reached Congress, the ground work had been done. It was received with enthusiasm on both sides of the aisle. On 6 February 1958, the Senate created the Special Committee on Space and Astronautics. In a statement confirming its priority, Lyndon B. Johnson, the Senate Majority Leader from Texas, was named chairman. The House mirrored the move when on 5 March, it created a Select Committee on Astronautics and Space Exploration, naming the Speaker of the House John W. McCormack from Massachusetts as the chairman. Hearings got underway in April and three months later on 16 July, Congress passed the National Aeronautics and Space Act of 1958 (Public Law 85D568) that NACA would become NASA after 90 days, unless the transition was proclaimed sooner by the new NASA administrator. Thus, less than six months after signing off on the proposal to Congress, the piece of legislation establishing the National Aeronautics and Space Administration as a legal United States government organization returned to Eisenhower's desk where it was signed on 29 July 1958.

Thus in one move, NACA and its 8,000 employees were absorbed intact into the new space agency. Its three research Centers (Ames Aeronautical Laboratory, Langley Aeronautical Laboratory and Lewis Flight Propulsion Laboratory), along with the Army's Jet Propulsion Laboratory, formed the space agency's first Field Centers.[50] NACA's meager $100 million annual operating budget would soon pale in draconian fashion to the funding that the new Agency would begin receiving as America committed itself to the space race.

On 1 October 1958, just three days shy of the one year anniversary of Sputnik, NASA officially opened it doors as the nation's space agency.